Advanced Composites and Hybrid Materials (2026) 9:150
https://doi.org/10.1007/s42114-026-01715-9

RESEARCH

Processing-induced magnetic softening in high-loading printable composites for additive manufacturing

Gloria Guerrero-Muñoz[1] · Jorge Revuelta-Losada[1] · Jia Yan Law[1] · Elisa Guisado-Arenas[1] · Luis M. Moreno-Ramírez[1,3] · Manuel Sanchez-Poncela[2] · Victorino Franco[1]

Received: 18 November 2025 / Revised: 26 January 2026 / Accepted: 20 February 2026
© The Author(s) 2026, modified publication 2026

Abstract
Additive manufacturing (AM) of magnetic composites has traditionally been limited by trade-offs between filler loading, filament quality, processability, and functional performance. Yet, processing-driven advances in AM can redefine this design space, enabling unprecedented filler loadings, broader material compatibility, and scalable fabrication beyond conventional flexible metastructures. Here, we introduce a versatile feedstock-engineering strategy that fabricates magnetic composite filaments with high filler contents while maintaining homogeneity, printability, and performance. By systematically comparing three polymer matrices with two classes of soft magnetic powders, we report the critical interplay among matrix chemistry, filler morphology, and processing conditions, achieving magnetic filaments at high loadings (up to 61 wt% in rigid and 52 wt% in flexible systems), well above commercial standards (~40 wt%). Polyethylene terephthalate yields rigid, defect-free filaments at high loadings, whereas thermoplastic polyurethane uniquely enables mechanically flexible, magnetically functional filaments. Structural analyses confirm uniform filler dispersion and preservation of the amorphous or crystalline characters of the powders, while magnetic characterization uncovers an unexpected processing-induced magnetic softening: extrusion reduces coercivity relative to raw powders. This counterintuitive softening transforms extrusion from a passive fabrication step into a functional design tool. Proof-of-concept flexible demonstrations, including flexible actuators and magnetic grippers, exhibit magnetic field-responsive deformation and flux-guided manipulation, validating multifunctionality. The results establish a new design principle, processing-induced property enhancement, which enables AM of magnetically active yet mechanically compliant and reconfigurable systems for soft robotics and electromagnetic control. This scalable, polymer-independent method expands material and functional options for engineering multifunctional composites with structural integrity, mechanical flexibility, and enhanced functionality.

Keywords Additive manufacturing · magnetic filaments · fused fabrication filament · soft magnetic properties · mechanical/secondary properties

1 Introduction

Additive manufacturing (AM) enables the fabrication of complex structures with integrated geometry and functionality, offering significant advantages for developing multifunctional composites. Within AM of functional systems, magnetic composites present unique opportunities to design devices with tailored magnetic responses, mechanical adaptability, and structural complexity, with applications ranging from soft robotics, field-responsive actuators, wireless power transfer, and magnetocaloric systems [1–8]. Among the AM techniques, fused filament fabrication (FFF) is particularly advantageous over stereolithography, direct ink writing or metal-based powder

✉ Jia Yan Law
jylaw@us.es

✉ Victorino Franco
vfranco@us.es

[1] Multidisciplinary Unit for Energy Science, Department of Condensed Matter Physics, Universidad de Sevilla, P.O. Box 1065, Sevilla 41080, Spain

[2] ArcelorMittal Global R&D, Calle Calafates 9, Avilés 33490, Asturias, Spain

[3] Departamento de Electricidad y Electrónica, Universidad de Valladolid, Valladolid 47011, Spain

Published online: 05 March 2026

bed fusion methods such as laser - powder bed fusion (L-PBF), electron beam - powder bed fusion (EB-PBF), owing to several key advantages: low equipment and operational costs, compatibility with a broad library of thermoplastic polymers, and ease of integrating composite filaments into complex 3D architectures [9–14]. Unlike L-PBF or EB-PBF, which require costly high-energy sources, controlled atmospheres, and specialized powders, FFF offers an affordable and scalable route accessible to small businesses, research institutions, and industrial users alike [14–16].

Despite these advantages, commercially available magnetic filaments are typically Fe-based, which limits performance due to relatively low filler contents of ~40 wt% [17, 18]. While alternative magnetic fillers can partially address these limitations, higher loadings often compromise dispersion homogeneity or induce extrusion clogging or require solvent-casting methods that are difficult to scale and restricted to specific chemistry-compatible recipes. Flexible magnetic composites have also been explored, often via solvent casting, with filler loadings remaining modest (<30 wt%) or not reported [19–23].

The increase of magnetic filler content in polymer filaments is consequently inherently limited by issues of processability and mechanical integrity. While rheological instability and filament brittleness typically onset beyond 40 wt%, these cumulative processing challenges have effectively confined the practical ceiling for soft magnetic composites to approximately 50 wt%. This limitation contrasts significantly with hard magnetic or magnetocaloric systems, which are frequently designed for much higher volumetric fractions to optimize energy product or cooling efficiency. For instance, literature reports exceptional loadings for hard magnetic systems, such as ~86 wt% (55 vol%) $SrFe_{12}O_{19}$ in PA12 [24] or highly filled Nd-Fe-B bonded magnets [25]. However, such systems maximize volumetric density to optimize coercivity and energy product, often relying on rigid matrices or fine nozzles (e.g., 0.3 mm [24]) that limit compliance or printer compatibility. This design philosophy contrasts with soft magnetic composites composites for actuation, which must strike a balance between magnetic response, mechanical compliance, and printability; excessive loading in these systems would drastically increase stiffness and hinder actuation capability.

Soft magnetic composites, which respond reversibly to applied magnetic fields, are particularly attractive for emerging applications, such as soft robotics and adaptive actuators that demand not only magnetic functionality but also additional mechanical flexibility. However, the fabrication of such magnetic composite filaments remains restricted by available feedstock options. Industrially prepared pellets offer limited compositions and filler concentrations [4, 26–28], constraining design freedom and material optimization. More critically, the development of high-loading soft magnetic filaments is challenged by the complex interplay between filler loading, polymer processability, and mechanical compliance. For instance, acrylonitrile butadiene styrene+steel composites exhibit severe reductions in strength and increased brittleness beyond critical filler fractions [29, 30], highlighting the difficulty of maintaining printability and flexibility at elevated magnetic loadings. Furthermore, thermal processing in FFF filament fabrication is generally regarded as a passive stage, primarily involving the transfer of properties from raw powders to the composite. This perspective neglects the inherent structural instabilities of functional fillers, notably amorphous soft magnetic powders, which are vulnerable to crystallization or oxidation during thermal processing as they are subjected to elevated extrusion or nozzle temperatures. Consequently, filament fabrication may be considered a potential drawback as it can lead to material degradation.

Here, we introduce a feedstock-engineering (encapsulation-extrusion) approach that enables scalable fabrication of high-loading soft magnetic filaments that can be rigid or mechanically compliant. This strategy conceptually differs from earlier approaches: in contrast to high-loading systems optimized for hard-magnetic coercivity [24, 25], or solvent-cast formats restricted to specific chemistry recipes [19–23], our focus is on soft-magnetic, mechanically compliant filaments engineered directly for FFF. We investigated three thermoplastic matrices, polylactic acid (PLA) and polyethylene terephthalate glycol (PETG), representing rigid systems, and thermoplastic polyurethane (TPU), a flexible polymer, and two soft magnetic fillers, M300 maraging steel and Fe-Nb-Si-B-Cu (rationale for material selection detailed in Supplementary Information). This approach achieves filler loadings up to 61 wt% in PLA, 52 wt% PETG, and 52 wt% in TPU, which surpass the filler fractions reported in commercial filaments (~40 wt%) and, for soft, flexible systems, represent the highest values documented in the literature [19–23]). Systematic studies across three polymer matrices and two classes of soft magnetic fillers reveal the critical interplay between matrix chemistry, filler morphology, and processing conditions, which governs printability, mechanical compliance, and magnetic performance. Remarkably, the composites retain their amorphous structure during processing and exhibit consistent magnetic softening: extrusion lowers coercivity relative to raw powders. The dual advance of combining such high, yet processable, filler loadings with enhanced soft magnetic properties establishes a new strategy for AM of functional, mechanically compliant magnetic composites. By transforming extrusion into a functional design tool, this approach enables 3D-printed magnetic composites that outperform their precursors in both processability and performance, establishing a new paradigm

in AM. The strategic methodology is fully compatible with thermoplastic extrusion, not confined to particular polymer chemistries, yet accommodating both rigid, high-loading filaments and surpassing commercial and solvent-cast analogues. This broadens the FFF material spectrum for multifunctional, reconfigurable magnetic devices.

2 Experimental

The starting materials of this work used commercial polymeric resins of PLA, modified PETG, and TPU (FLEX 98 A) from *Smart Materials 3D*. The magnetic fillers were *AdamIQ* M300 maraging steel [31, 32] and Fe-Nb-Si-B-Cu bulk metallic glass (BMG) powders fabricated by *Arcelor-Mittal*. Their measured compositions are listed in Table S1, Supplementary Information.

Composite and neat filaments were produced using a single-screw filament extruder (*3Devo* B.V., The Netherlands). The latter was fabricated to serve as a reference throughout this work and also to print the neat capsules for composite feedstock preparation. The polymer loss during the feedstock preparation was minimal (<0.9 wt%, Table S2), ensuring nominal compositions as listed in Table 1. The feedstock engineering strategy developed to achieve these nominal compositions is described in detail in the following section.

Extrusion temperatures and screw speed profiles of the filament fabrication were optimized depending on composition (detailed in later sections) to account for their differing rheological behaviors. The extruded filaments were monitored in real time to maintain a consistent diameter of 1.75 ± 0.05 mm, ensuring FFF compatibility.

Thermogravimetric analysis (TGA; TGA Discovery) was performed under a nitrogen atmosphere. Samples of ~10 mg were heated from room temperature to 850 °C at a rate of 10 °C min^{-1}. Weight loss data were used to determine actual filler loadings and to evaluate polymer thermal stability. Cross-sections of extruded filaments were prepared by mechanical cutting and examined by field-emission scanning electron microscopy (FESEM, *Zeiss Teneo*) at an accelerating voltage up to 30 kV. Backscattered electron (BSE) imaging was also used to evaluate filler dispersion and distribution. Phase analysis was carried out using an x-ray diffractometer (Bruker D8 Advance, Cu K_α radiation, λ = 1.5406 Å). Data were collected in the 2θ range of 25–120 º. Patterns from composite filament were compared with those of the raw powder to confirm structural stability and amorphous character.

The mechanical properties of the composite filaments were evaluated through a quantitative flexibility test. Unlike standard structural characterization methods such as ASTM D790 (Standard Test Methods for Flexural Properties) [33] or ASTM D695 (Standard Test Method for Compressive

Table 1 Target and achieved compositions of magnetic composite feedstock prepared using the encapsulated feedstock engineering strategy. T_s = sealing temperature; †Sealing loss refers to the polymer mass loss during feedstock preparation and was determined by the difference in polymer mass before and after filler loading. ‡To enable direct comparison with literature, the filler loadings were expressed in both weight% (wt%) and converted volume percent (vol%) using standard mass–volume conversion relations (Eq. S1) based on measured material densities (densities of PLA, PETG, TPU, M300 maraging steel and Fe-Nb-Si-B-Cu = 1.24, 1.27, 1.09, 8.6 and 7.9 g cm^{-3}, respectively)

Filament ID	Polymer matrix	Magnetic filler	Target loading (wt%)	Measured Loading ± Standard Deviation (wt%)	Calculated Loading (vol%)	T_s (ºC)	Sealing loss (wt%) †	Filament outcome	Highlights
PLA + M300 maraging	PLA	M300 maraging steel	65	67.1 ± 0.6	23	230	0.40	Continuous, rigid filament	Highest filler loading; smooth extrusion; brittle under bending
PETG + M300 maraging steel	PETG	M300 maraging steel	55	57 ± 3	17	260	0.37	Continuous, rigid filament	Smooth extrusion; shiny surface
TPU + M300 maraging steel	TPU	M300 maraging steel	55	56.3 ± 1.5	17	220	0.80	Continuous, flexible filament	Matte surface; mechanically compliant; repeatedly magnetically responsive
PETG + Fe-Nb-Si-B-Cu	PETG	Fe-Nb-Si-B-Cu	50	47 ± 3	14	260	0.82	Continuous, flexible filament	Smooth extrusion; shiny surface; demonstrates processing-induced coercivity softening; retained amorphous phase

Properties) [34], which require molded or printed coupons of specific geometries, this protocol was designed to directly assess the 'spoolability' and feedstock integrity of the raw cylindrical filaments. In this setup, a 5-cm-long filament sample was placed beside a ruler and gradually pushed against a fixed wall using a movable transparent block until fracture occurred. Displacement was measured from the initial sample length (X_0) to the block position at breakage (X_f), and flexibility was expressed as the displacement-to-length ratio, $r = (X_0 - X_f) / X_0$. This metric r serves as a normalized proxy for the minimum bend radius the feedstock can withstand during 3D printing (e.g., passing through guide tubes and extruder). Higher values of r indicate greater deformability, while samples that did not fracture were considered fully flexible ($r \sim 1$). Each filament type was tested in quintuplicate, and the results are summarized in Table S3. This straightforward method provides a reproducible metric to distinguish between brittle feedstocks (prone to fracture during spooling) and compliant composites suitable for flexible applications.

Magnetic properties were measured using a vibrating sample magnetometer (VSM 8607, LakeShore Cryotronics Inc.) at room temperature. Powder samples were measured in green compact form, while filament samples were cut into segments of length 0.05 mm. Magnetization (M) versus magnetic field (H) was recorded up to a maximum applied field of 2.2 T. Coercivity (H_c), and saturation magnetization (M_s) were extracted for comparison across compositions. The latter was also used for determining the magnetic filler loadings in the composite filaments and compared to results from TGA. All printing experiments were performed on a fused filament fabrication printer (*Prusa MK3S+*). Printing parameters (nozzle temperature, bed temperature, print speed, and cooling) were optimized for each composition and are listed in later sections. Rigid composite filaments were printed into structural test geometries, including 3D cubes and snowflake architectures, whereas flexible composites were demonstrated for their capability in coiling, twisting, bending, and magnetically actuation.

3 Results and discussion

3.1 Fabrication of high-loading soft magnetic composite filaments with high filler loadings

Conventional feedstock preparation strategies for magnetic composites are inherently limited. Solvent casting approaches, while effective for filler dispersion, are constrained by chemical compatibility, scalability challenges, and typically restrict filler loadings to below 30 wt% in flexible matrices. Similarly, commercial magnetic filaments rarely exceed 40 wt% fillers [17, 18] as higher loadings compromise extrusion continuity and mechanical performance. To overcome these constraints, we developed a versatile feedstock-engineering strategy that enables scalable fabrication of continuous magnetic composite filaments at high filler loadings compatible with FFF (Fig. 1).

3.1.1 Feedstock engineering strategy

In contrast to conventional solvent-based or matrix-specific routes, this approach employs a polymer-independent encapsulation technique. As illustrated in Fig. 1a, magnetic powder is loaded into 3D-printed polymer capsule arrays and hermetically sealed by localized thermal bonding. This engineered encapsulation strategy addresses two critical failure modes of high-loading composites: it eliminates air gaps to void formation and hermetically seals the powder to prevent detachment. Mechanistically, this strategy creates a "pre-wetted" macro-composite; the polymer capsule walls act as a lubricating interface during the initial stages of extrusion, effectively replacing high-friction particle-particle interactions with polymer-polymer friction. This allows for stable flow even at volumetric fractions that would typically seize a standard twin-screw compounder.

3.1.2 Compositional integrity and validation

To ensure precise filler loading, a rigorous gravimetric protocol was employed. After establishing the sealing temperature (T_s, detailed in Table 1) for each system, the nominal feedstock composition was determined gravimetrically. The polymer capsule arrays and lids were first weighed prior to encapsulation and then re-weighed after loading with magnetic fillers and sealing, allowing the mass of encapsulated fillers to be obtained by difference. Any capsules that failed the sealing test (i.e., showed powder leakage upon mechanical handling or impact) were discarded before this mass measurement. As a result, the reported feedstock composition, and thus the steady-state filament collected for spooling and subsequent printing, originates exclusively from capsules sealed at the validated T_s and exhibits a well-defined composition. Quantitative composition analysis (Table 1) confirms high filler loadings of 67.1 wt% in PLA, 57.3 wt% in PETG, and 56.3 wt% in TPU, surpassing commercial magnetic filaments (~40 wt% Fe [17, 18]). The minor polymer loss during capsule sealing (< 0.9 wt%, Table S2) demonstrates the effectiveness of the hermetic sealing step, ensuring that the prepared feedstock retains its designed composition prior to extrusion.

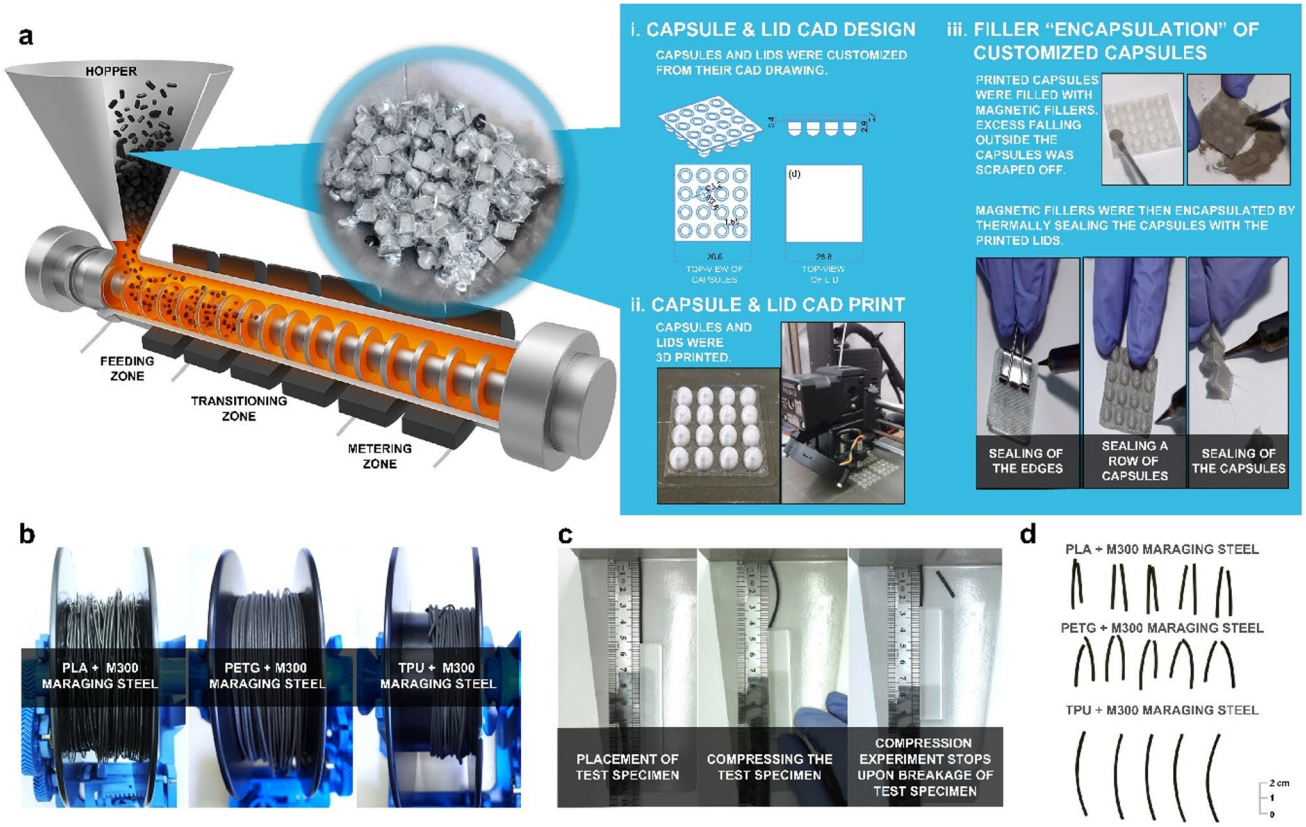


Fig. 1 Solvent-free feedstock engineering for scalable fabrication of homogeneous magnetic filaments. (**a**) Illustration and photographs of the feedstock engineering strategy: encapsulation of magnetic fillers within polymeric capsules, followed by controlled hermetically sealing before controlled thermal extrusion into continuous composite filaments. Encapsulation geometry: capsules (outer diameter 5.2 mm, inner diameter 3.6 mm, 0.8 mm wall thickness) in 28.8 mm × 28.8 mm array; sealing lids 28.8 mm × 28.8 mm × 4 mm. Sealing temperatures are detailed in Table 1. (**b**) Photographs showing spools of extruded composite filaments. (**c**) Top view of flexibility test measurements. The test specimen was placed in line with a transparent block, which was used for compressing the test specimen. (**d**) Flexibility demonstrations showing that TPU composites retain integrity under bending and compression, whereas PLA- and PETG composites fracture under similar tests. The approach enables scalable, solvent-free, and polymer-independent production of high-loading magnetic filaments with excellent homogeneity and printability

3.1.3 Process optimization and extrusion behavior

The encapsulated feedstocks were extruded into filaments using optimized parameters summarized in Table 2. Residence time within the extruder, cooling rate of the extruded filaments, and extrusion screw speed were independently tuned for each polymer and its magnetic composites. Higher screw speeds shorten residence time within the extruder, resulting in more solidified extrudate that, in turn, requires lower fan settings to achieve continuous spooling [35]. Distinct processing windows observed for PLA, PETG, and TPU filaments correlate with their chain chemistry and thermophysical properties (glass transition temperatures and melting point) in agreement with literature [36–38]. PLA, with its relatively low melting point, requires the lowest extrusion temperatures, whereas PETG demands higher temperatures due to its strength and thermal stability [38]. TPU, by contrast, requires intermediate-to-high extrusion temperatures to overcome chain flexibility and mobility. The incorporation of fillers substantially altered the flow behavior. While low particle loadings can facilitate flow via a "ball-bearing" effect [4], this lubrication regime disappears at the high loadings targeted here. Instead, flow resistance increases markedly. To maintain stable melt rheology and prevent nozzle clogging, extrusion temperatures were elevated relative to neat polymer baselines. This behavior is consistent with the recent studies highlighting the nonlinear rheological signatures of highly filled polymer suspensions. Importantly, despite the increased resistance, the "encapsulation" mechanism allowed for continuous spooling without filament breakage, a common failure point in conventional highly-filled systems.

Table 2 Optimized extrusion parameters for fabricating neat and magnetic composite filaments. Processing temperatures and screw speeds were tuned to balance viscosity and homogeneity, defining matrix-specific operational windows. Temperatures of the four heating zones in the extruder are also included

Filament ID	Polymer matrix	Nominal filler loading (wt%)	Drying phase	Feeding zone	Transitioning zone		Metering zone	Screw speed (rpm)	Fan settings (%)
				T_{feed} (°C)	T_{tran_start} (°C)	T_{tran_end} (°C)	$T_{metering}$ (°C)		
PLA	PLA	-	45 °C, 8 h	170	185	190	180	3.5	80
PETG	PETG	-	65 °C, 8 h	219	228	232	220	5.0	30
TPU	TPU	-	55 °C, 8 h	173	176	165	163	4.0	80
PLA + M300 maraging steel	PLA	67.1	45 °C, 8 h	175	185	190	186	4.0	60
PETG + M300 maraging steel	PETG	57.3	65 °C, 8 h	219	228	232	220	5.0	30
TPU + M300 maraging steel	TPU	56.3	55 °C, 8 h	194	202	198	192	5.2	35
PETG + Fe-Nb-Si-B-Cu	PETG	47.4	65 °C, 8 h	221	226	234	227	5.0	20

Table 3 Comparison of filler loadings in reported magnetic polymer filaments suitable for FFF printing. †Hard magnetic and magnetocaloric examples included for context but are not directly comparable due to differing magnetic response and processing requirements. Note that our works use a larger nozzle size (0.8 mm, 205–255 °C), enabling soft magnetic response with mechanical flexibility and broader printer compatibility in comparison to the record loading of $SrFe_{12}O_{19}$ hard magnetic composite filaments (0.3 mm, 260 °C) [24]

System types	Filler/Material	Wt.% fillers	Vol.% fillers (estimated /reported)	Printability & notes	Reference
Soft magnetic rigid	PLA–M300	67.1 ± 0.6	23	Excellent FFF, rigid, printable (0.8 mm nozzle, 205 °C)	This work
	Fe-PLA (commercial)	~40	~8–10	Good FFF compatibility, rigid	[17, 18]
	Steel-PLA	30–52	~8–15	Good extrusion, brittle	[3, 4]
	Ferrite-PLA	< 20	< 5	Could clog at higher loadings, rigid	[44]
Soft magnetic flexible	Various solvent-cast	< 30	< 8	Not filament-extrudable	[19–23, 39]
	PETG–M300	57 ± 3	17	Excellent FFF, flexible, printable (0.8 mm nozzle, 240 °C)	This work
	TPU–M300	56.3 ± 1.5	17	Excellent FFF, flexible, printable (0.8 mm nozzle, 250–255 °C)	This work
	PETG–FeNbSiBCu	47 ± 3	14	Excellent FFF, flexible, printable (0.8 mm nozzle, 240 °C)	This work
Hard magnetic rigid	$SrFe_{12}O_{19}$-polyamide	~86	~40–55	Printable (0.3 mm nozzle, 260 °C)	[24]
Hard magnetic flexible	Nd-Fe-B-flexible polymer (polyether ether ketone)	Up to 75	~40	Degraded printability with higher loadings, semi-flex	[40]
Hard magnetic	$Nd_2Fe_{14}B$-ECOFLEX (soft)	6:4 mass ratio of Nd-Fe-B: ECOFLEX	Cannot be estimated as it is cast into a hollow sphere using a spherical mold	Hollow molded spheres, not FFF filament	[41]

3.1.4 Filament flexibility and benchmark comparison

Representative spooled filaments and flexibility tests are shown in Fig. 1b. These results highlight the mechanicality of the matrices: while the TPU matrix did not break during flexibility bending tests, the rigid PLA and PETG systems fractured under similar strains. Importantly, despite these differences in ultimate flexibility, all produced systems possessed sufficient mechanical integrity to be continuously spooled onto standard reels.

To contextualize the filler loadings, Table 3 compares the achieved loadings with literature benchmarks. Both weight and volume fractions are presented to enable density-independent benchmarking. For instance, while hard magnetic $SrFe_{12}O_{19}$-PA12 filaments achieve exceptional 86 wt% (~55 vol%) loading, a nozzle size of 0.3 mm and printing temperature of 260 °C are optimized for high coercivity in their rigid matrices. In contrast, soft magnetic applications require a balance of magnetic response and mechanical compliance. Our results in flexible TPU represent one of the highest

processable loadings for soft magnetic, compliant filaments. Unlike the rigid hard magnetic benchmarks, our system utilizes a larger printing nozzle (0.8 mm) to preserve the flexible mechanics required for soft actuation. Furthermore, the larger nozzle size enables broader printer compatibility. These distinctions are particularly evident for flexible systems, where literature precedents remain scarce or rely on non-extrudable processing routes. Collectively, these results establish a generalizable feedstock-engineering framework for producing homogeneous, extrusion-compatible magnetic composites, a scalable route for expanding the FFF materials library toward functional, high-loading soft magnetic systems.

Photographs of the tested specimens from flexibility bending demonstrations are shown in Fig. 1c, with mechanical flexibility data summarized in Table S3. PLA- and PETG-based composites fracture under bending tests due to their inherent rigidity with added filler-induced brittleness. In contrast, TPU-based composites retain remarkable compliance despite filler loadings, sustaining repeated bending cycles without fracture (Fig. 1d). The success of the versatile fabrication strategy of these composites rests on two critical factors: (i) hermetic encapsulation eliminating air gaps, and (ii) extrusion-process optimization tailored to polymer chemistry and rheological transformations at high filler concentrations. Together, these advances overcome long-standing limitations in AM of magnetic composites and offer a versatile route to functional, high-performance printing materials.

3.2 Microstructural uniformity and compositional determination of M300 maraging steel composites

Electron micrographs of M300 maraging steel composite filaments confirm uniform dispersion of M300 maraging steel particles across all polymer matrices, even at maximum loadings, with no observation of agglomeration across filament cross-sections (Fig. 2a). The particles are predominantly spheroidal, 40–100 μm in diameter, although some appear irregular due to surface removal during polishing. Whole cross-sections (see insets) confirm that the apparent voids originate from particles lost during cutting and polishing (particle pull-out), rather than from intrinsic pores or smaller unresolved fillers. Matrix-specific differences are evident: PLA composites display the highest particle density, consistent with their greater nominal volume fraction; PETG composites exhibit more

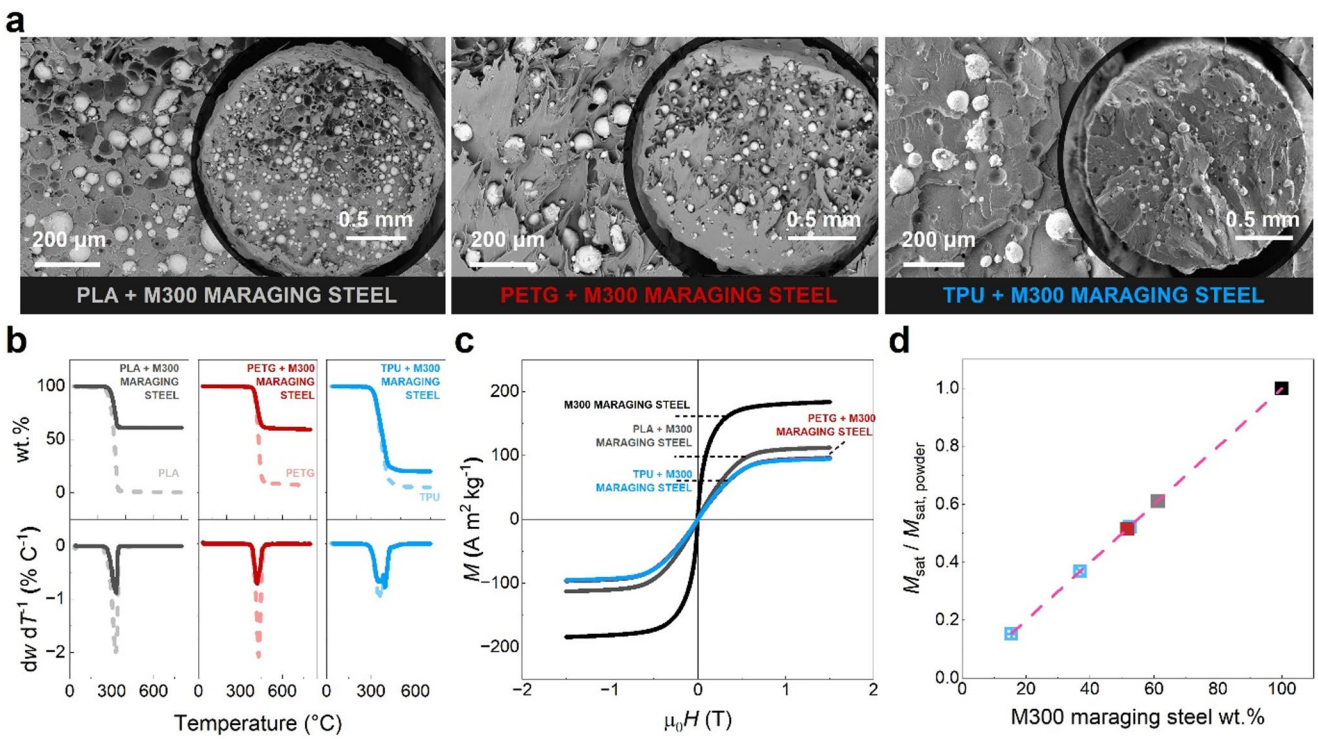


Fig. 2 High-loading composites show uniform filler distribution and phase stability with magnetic response that scales with filler loading. (**a**) Microstructures of cross-sections show uniform filler distribution in the M300 maraging steel composites without agglomeration. Insets (right) of full view of the cut cross-sections reveal that surface voids arise from particle loss during cutting, not intrinsic porosity. (**b**) Thermogravimetric analysis indicates preserved thermal stability. (**c**) Magnetic measurements of raw powder and composites show soft magnetic behavior. (**d**) Normalized saturation magnetization ($M_{sat}/Msat,_{powder}$) ratio scales linearly with filler content (determined from magnetic) across multiple samples, demonstrating the reliability of magnetometry for compositional analysis. Measurements of the various pieces of TPU + M300 maraging steel plotted here were taken from the start to the end of extrusion. Their results indicate that the composite filament contains up to 52 wt% fillers, which is consistent with the nominal composition of the feedstock

deeply embedded particles; and TPU composites initially show fewer visible fillers (compared to their nominal values), as their samples were taken from the beginning of extrusion when excess polymer remained in the barrel.

Thermogravimetric analysis (TGA) demonstrates that the filler incorporation did not compromise the thermal stability of the composites. The onset degradation temperatures remain high at 330, 420, and 430 °C for PLA-, PETG-, and TPU-based M300 maraging steel composites, respectively (Fig. 2b). For PLA, degradation occurred around 330 °C, leaving a small ash residue of 0.1%. The PLA composite with 67.1 wt% nominal filler exhibits a slightly lower experimental fraction of 60.8 wt%, likely due to the variability during the filament fabrication. PETG degrades at approximately 420 °C with a 7.1% residue, and its 57.3 wt% composite yields 59.4 wt%, closely matching the nominal value. TPU shows degradation onset before 430 °C, leaving a 4.9% residue. The TPU + M300 maraging steel composite sampled for TGA showed a filler content of 19.9 wt%, lower than the nominal 56.3 wt% feedstock input. This discrepancy reflects the rheological stabilization lag specific to the elastomeric matrix; the compliant TPU capsules undergo compression in the extruder during startup, resulting in a transient polymer-rich flow. However, this value is not representative of the steady-state spooled filament. As discussed in the following paragraph, magnetometry performed on the spooled filament confirms a consistent filler fraction of 52 wt%, aligning with the nominal loading. Derivative thermograms in the bottom panel of Fig. 2b show that the maximum rate of mass loss occurs at similar temperatures for PLA and PETG composites, with minor shifts to lower temperatures indicating that the degradation mechanism is largely unaffected by filler addition, consistent with ref [4]. The TPU composite exhibits a less pronounced peak, directly attributed to the lower filler concentration in this specific transient sample rather than an intrinsic reduction in thermal stability. Overall, these results confirm that M300 maraging steel fillers do not significantly alter the thermal decomposition behavior of the polymer matrices.

Magnetic measurements (magnetization, M, versus applied magnetic field, H) of the raw powder and composite filaments confirm that the latter preserves the soft magnetic character of the raw M300 maraging steel powder, with low coercivity, H_c, retained across all systems (Fig. 2c). Crucially, this magnetization data serves as a robust probe for bulk homogeneity along an extruded filament, given that all fillers contribute to the magnetic signal. As anticipated, saturation magnetization scales with filler content, reflecting dilution by the non-magnetic polymer matrices (Fig. 2d). A universal linear correlation ($R^2 \approx 1$) is established between the normalized saturation magnetization and the filler fraction, anchoring the composites to the 100% pure powder reference. Quantitative analysis further confirms that the actual filler loadings closely match the nominal feedstock compositions, yielding 61, 52, and 52 wt% for PLA + M300 maraging steel, PETG + M300 maraging steel, and TPU + M300 maraging steel composites, respectively. The fact that the TPU composite data points align perfectly with the rigid PLA- and PETG-M300 controls, which are free from elastic feeding artifacts, confirms that the low loading observed in TGA sampling was a start-up artifact, and the TPU composite filament possesses a bulk homogeneity consistent with the nominal composition.

3.3 Filament fabrication processing-induced magnetic softening

Among the three polymer matrices investigated, PETG enabled the most straightforward fabrication of continuous, homogeneous composite filaments. PETG was therefore selected to incorporate gas-atomized Fe-based BMG alloy powders (Fe-Nb-Si-B-Cu) at a nominal 47 wt% loading (calculated from the customized feedstock as presented in Table 1). Its filament fabrication parameters are summarized in Table 2. The resulting spooled filaments are shown in Fig. 3a. X-ray diffraction (XRD) shows a broad hump without any well-defined crystalline peaks, confirming that the amorphous character of Fe-Nb-Si-B-Cu powder is fully preserved after melt compounding and filament extrusion (Fig. 3b). No crystallization or phase transformation was detected, demonstrating that the thermal processing maintains structural and compositional stability. The XRD of M300 maraging steel powder can be referred to Fig. S1.

TGA analysis further reveals that Fe-Nb-Si-B-Cu was incorporated at a filler content exceeding the nominal value. At temperatures above the PETG degradation point (~420 °C), 52.0 wt% of the mass remained, indicating that higher-than-expected loadings are achievable without compromising phase stability (Fig. 3c). Flexibility testing reveals distinct differences between the PETG composites (Fig. 3d). While the PETG + Fe-Nb-Si-B-Cu filaments withstand repeated bending without fracture, the PETG + M300 maraging steel filaments failed brittlely under similar conditions. This contrasting behavior allows for an interpretation of the failure mechanism supported by recent literature on filled FFF composites [42]. The coarse M300 particles (40–100 μm) act as significant stress concentrators and large critical flaws that disrupt the continuity of the rigid PETG matrix. Conversely, the Fe-Nb-Si-B-Cu is characterized by a significantly finer particle size (2–12 μm), which provides a larger surface-area-to-volume ratio. The finer morphology facilitates better interfacial interaction and enhances load transfer, resulting in a more uniform stress distribution that mitigates the localized stress spikes responsible for fracture. BSE micrographs (Fig. 3e) display uniform filler dispersion for the fine-particle system (see insets

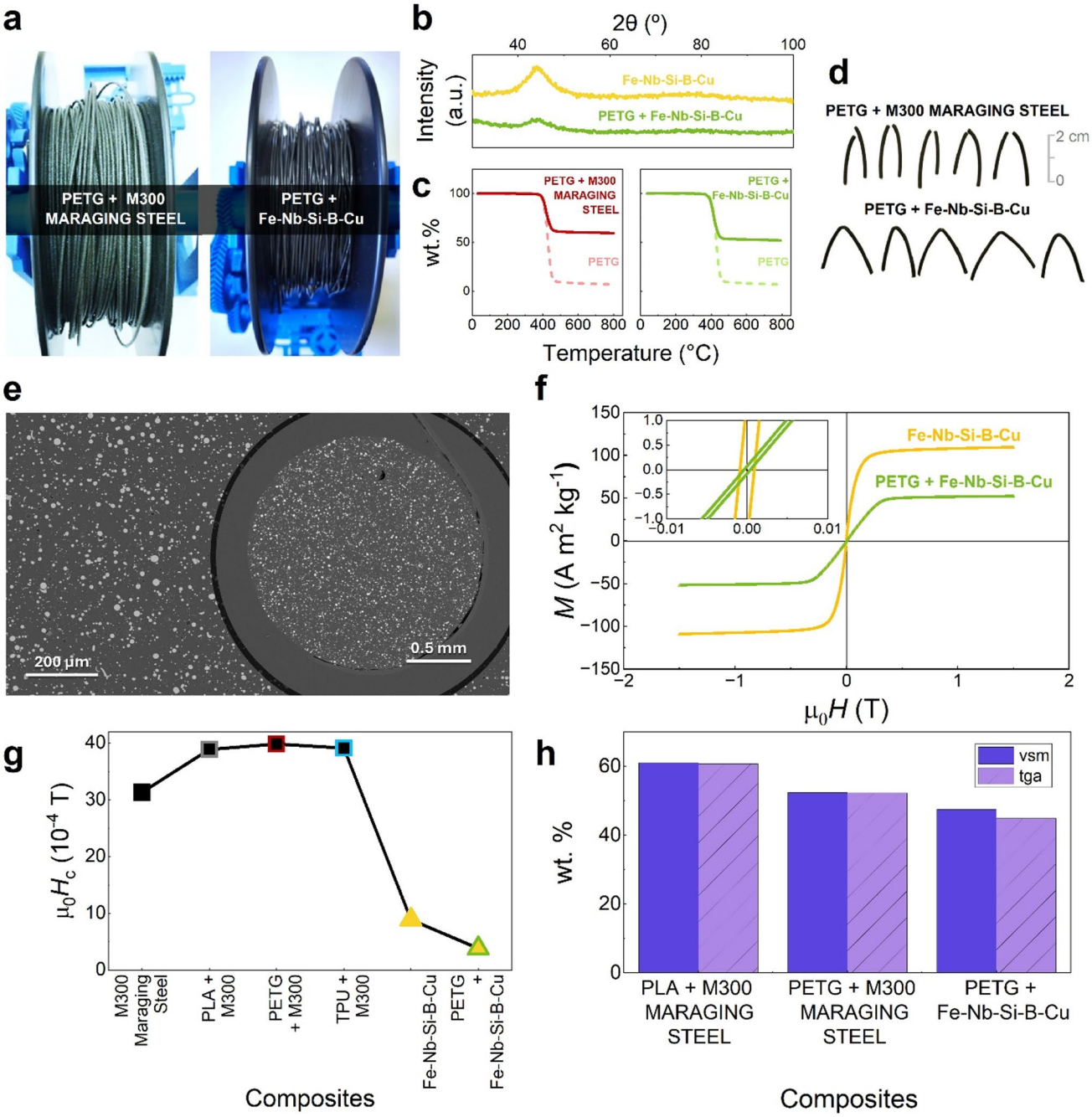


Fig. 3 PETG composites with Fe-Nb-Si-B-Cu fillers: phase stability and processing-induced magnetic softening. (**a**) Photographs of spooled M300 maraging steel and Fe-Nb-Si-B-Cu composite filaments. (**b**) XRD data showing that Fe-Nb-Si-B-Cu composite filaments retain their amorphous character after extrusion. (**c**) TGA analysis confirms high filler loadings in both PETG + M300 maraging steel and Fe-Nb-Si-B-Cu composite systems. (**d**) Photographs after flexibility tests comparing PETG + Fe-Nb-Si-B-Cu (no fracture) versus PETG + M300 maraging steel (fracture). (**e**) BSE micrographs of PETG + Fe-Nb-Si-B-Cu, with inset full cross-section showing uniform distribution of spheroidal particles (2–12 µm). (**f**) Magnetic measurements of raw Fe-Nb-Si-B-Cu powder vs. corresponding magnetic composite filaments, confirming soft magnetic character. Inset: Magnified plot showing the coercivity reduction ("magnetically softer") of Fe-Nb-Si-B-Cu composites. (**g**) Coercivity comparison of magnetic fillers and their composite samples, highlighting that processing emerges as a design lever that enhances processing-induced magnetic softening, as observed in PETG + Fe-Nb-Si-B-Cu. (**h**) Filler content determined by TGA and magnetometry, showing good agreement between methods

for complete cross-section views). Although occasional voids are observed, they appear isolated and are attributed to minor extrusion instabilities (like improper melting or thermal degradation of the filament, fluctuations in pressure, or air entrapment) rather than moisture contamination as all feedstock was dried prior to fabrication.

Magnetic measurements of Fe-Nb-Si-B-Cu powder and its filament reveal that the latter retains the intrinsic soft magnetic character of the filler, with saturation magnetization scaling linearly with loading (Fig. 3f). Unexpectedly, coercivity (H_c) decreases in the composites relative to the raw powders (see magnified plot in inset), indicating a "processing-induced magnetic softening". This effect is observed in PETG + Fe-Nb-Si-B-Cu among all studied samples (Fig. 3g), yielding composites that are magnetically softer than both the raw powder and the M300 maraging steel. We ascribe this reduction to the thermal history imposed during extrusion, interpreting it through the established microstructural evolution of Fe-based amorphous alloys. As reported in ref [43], alloys of this type typically undergo three distinct stages of evolution upon heat treatment which dictate magnetic performance: (1) Stress relaxation (Stage 1), occurring at low annealing temperatures (typically < 400 °C), which induces magnetic softening due to the reduction of magnetoelastic anisotropy, without initiating crystallization; (2) Nanocrystallization (Stage 2), occurring near 500 °C, where coercivity initially increases due to the onset of nanocrystallization and subsequently may decrease due to an increase of exchange coupling among nanocrystals due to the increasing volume fraction; and (3) Secondary Crystallization (Stage 3) involving the formation of boride-type phases at higher temperatures, which causes a significant increase in coercivity. It is worth noting that the processing-induced softening should happen for magnetic powders for which the softening associated to stress relaxation occurs in the temperature range at which the filament is manufactured or processed. Given that the extrusion process employed here operates at 186–227 °C, which falls squarely within the Stage 1 regime, we performed a control experiment to confirm that thermal cycle mimicking the extrusion profile (heated to 230 °C). Comparison of the hysteresis loops before and after heating (Fig. S2) reveals that this thermal treatment alone reduces the coercivity from 1 to 0.65 mT. This directly mirrors the reduced coercivity observed in the composite compared to powder in Fig. 3g. Crucially, this result demonstrates that FFF extrusion can serve as an active functional tuning step. In addition, comparison of filler fractions determined by TGA and magnetometry shows excellent agreement (Fig. 3h). Minor discrepancies relative to nominal compositions are attributed to the manufacturing tolerances of the feedstock capsules rather than processing losses.

3.4 3D printing demonstrations and performance benchmarking

Even neat polymer filaments require precise optimization of printing conditions to ensure proper layer adhesion and defect-free structures, especially the adhesion between the initial layers and the build plate, which heavily depends on the printing bed temperature. Incorporating magnetic fillers increases these challenges: viscosity rises, flow behavior shifts, and standard settings often fail to ensure reliable printing. To determine optimal printing conditions, the printability of the developed magnetic composite filaments was systematically tested using standard FFF printers under various nozzle and bed temperatures, as well as different printing speeds. Because of the high filler content, these composites need a carefully balanced thermal regime to maintain continuous extrusion and prevent nozzle clogging. Fig. 4a shows a process map linking nozzle temperature, bed temperature, and printing speed with observed print quality. The map highlights a specific operational sweet spot (indicated by the largest symbols) where the neat and composite filaments produce smooth layer deposition. The continuous, stable flow observed within this optimized window demonstrates that the printer nozzle diameter (0.8 mm) is sufficient to accommodate the large M300 particles (40–100 μm). Specifically, the characteristic size ratio of at least one order of magnitude between the particle and the nozzle, combined with the spheroidal powder morphology and high molten matrix viscosity, effectively prevent mechanical sieving or clogging. Consequently, long-term stability is maintained within the identified windows, with no observed nozzle clogging or flow interruptions. Beyond the sweet spot regime, excessive nozzle temperatures cause filament melt overflow and warping, while too low temperatures or very fast printing lead to under-extrusion and weak interlayer adhesion. The optimized printing parameters for PLA + M300 maraging steel composite filaments require a T_{nozzle} of 205 °C, which is higher than values reported for PLA composite filaments in literature [3, 5, 44]. This increase can be attributed to the combination of higher filler loading and the corresponding rise in melt viscosity. As reported in ref [45], at elevated filler fractions, particle-particle interactions contribute to the overall viscosity and reduce the sensitivity of flow behavior to temperature changes. As a result, a higher printing temperature is required to achieve a smooth surface of the printed part and complete interlayer welding during deposition.

Representative printed structures used for printing optimization are shown in Fig. 4b, demonstrated with the PLA+M300 maraging steel composite. These results confirm that even highly loaded filaments (> 60 wt%) can be reliably printed into complex 3D geometries without

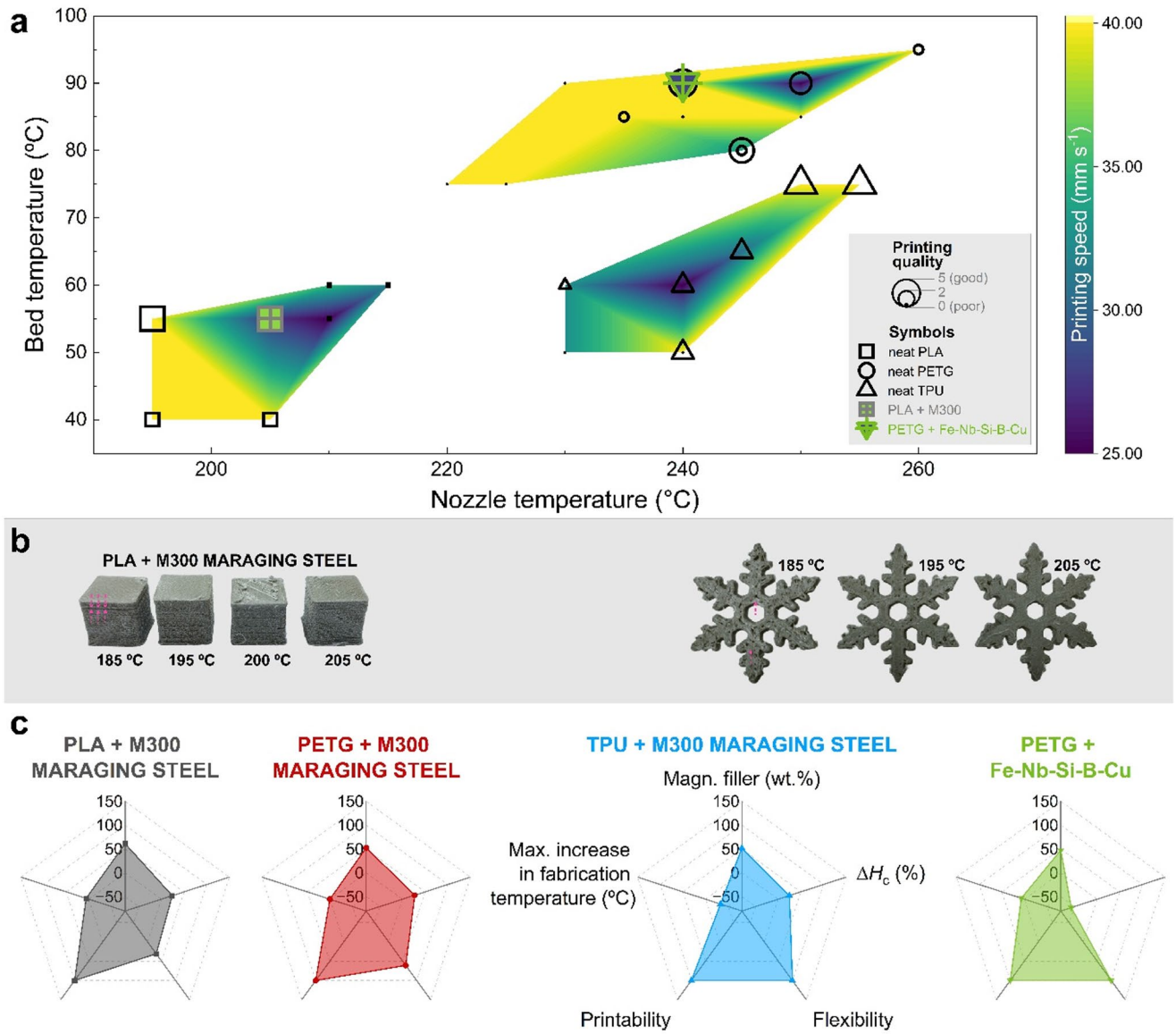


Fig. 4 Printability and comparative benchmarking of composite filaments. (**a**) 4D plot of the processing map illustrating the combined effect of nozzle temperature, bed temperature, and printing speed on print quality. The printing quality is represented on the scale of 0–5 for poor to good quality, represented by the symbol size for the different filaments. The largest symbol size highlights the optimal processing window corresponding to defect-free and dimensionally stable prints for neat and composite filaments. The optimized parameters for PLA+M300 and PETG-Fe-Nb-Si-B-Cu composite filaments are as follows: T_{nozzle} = 205 °C, T_{bed} = 55 °C, and printing speed=35 mm s^{-1}; and T_{nozzle} = 240 °C, T_{bed} = 90 °C, and printing speed=28 mm s^{-1}, respectively. (**b**) 3D printed structures from rigid composites (PLA+M300 maraging steel). Arrows indicate macroscopic defects (e.g., under-extrusion, warping) observed outside the stable processing window. In contrast, parts produced within the optimized regime exhibit consistent layer deposition, confirming that stable printing, i.e., free from mechanical sieving or bridging effects, is achievable despite the high particle loading. (**c**) Radar plots benchmark the different composite systems across multiple metrics, highlighting PETG+Fe-Nb-Si-B-Cu as a standout candidate for soft magnetic applications

fracture or nozzle clogging. The 5 cm x 5 cm x 5 cm cube and snowflake architectures indicate structural integrity achieved within the optimized printing regime. At 185 °C, the standard settings for neat PLA, poor melt flow of the composite causes interlayer gaps and voids (indicated by pink dashed arrows), reflecting incomplete layer fusion. In contrast, printing at 200–205 °C effectively suppresses these defects, producing denser and continuous structures with more uniform layer bonding. These architectures thus directly visualize the optimization of processing parameters for high-loading PLA+M300 maraging steel composites. Radar plots in Fig. 4c benchmark the composite systems across five key metrics: magnetic filler loading (wt%), coercivity variation (%), flexibility (%), printability

(%), and maximum temperature increase during processing (°C). Among all systems, PETG+Fe-Nb-Si-B-Cu proves to be especially promising, combining high filler incorporation with induced magnetic softening, mechanical resilience, and consistent printability (see Sect. 3.5). Overall, this process-property mapping offers a useful framework for additive manufacturing of high-loading soft magnetic composites and shows their compatibility with conventional FFF setups.

3.5 Functional demonstration

Building on the results from the structural and magnetic analyses, the functional application of the soft magnetic composites was experimentally demonstrated through magnetic actuation and flux-guided manipulation. The flexible TPU+M300 maraging steel filament was fixed between two poles and then subjected to an external magnetic field (Fig. 5a, top panel). It exhibits clear, reversible deflection

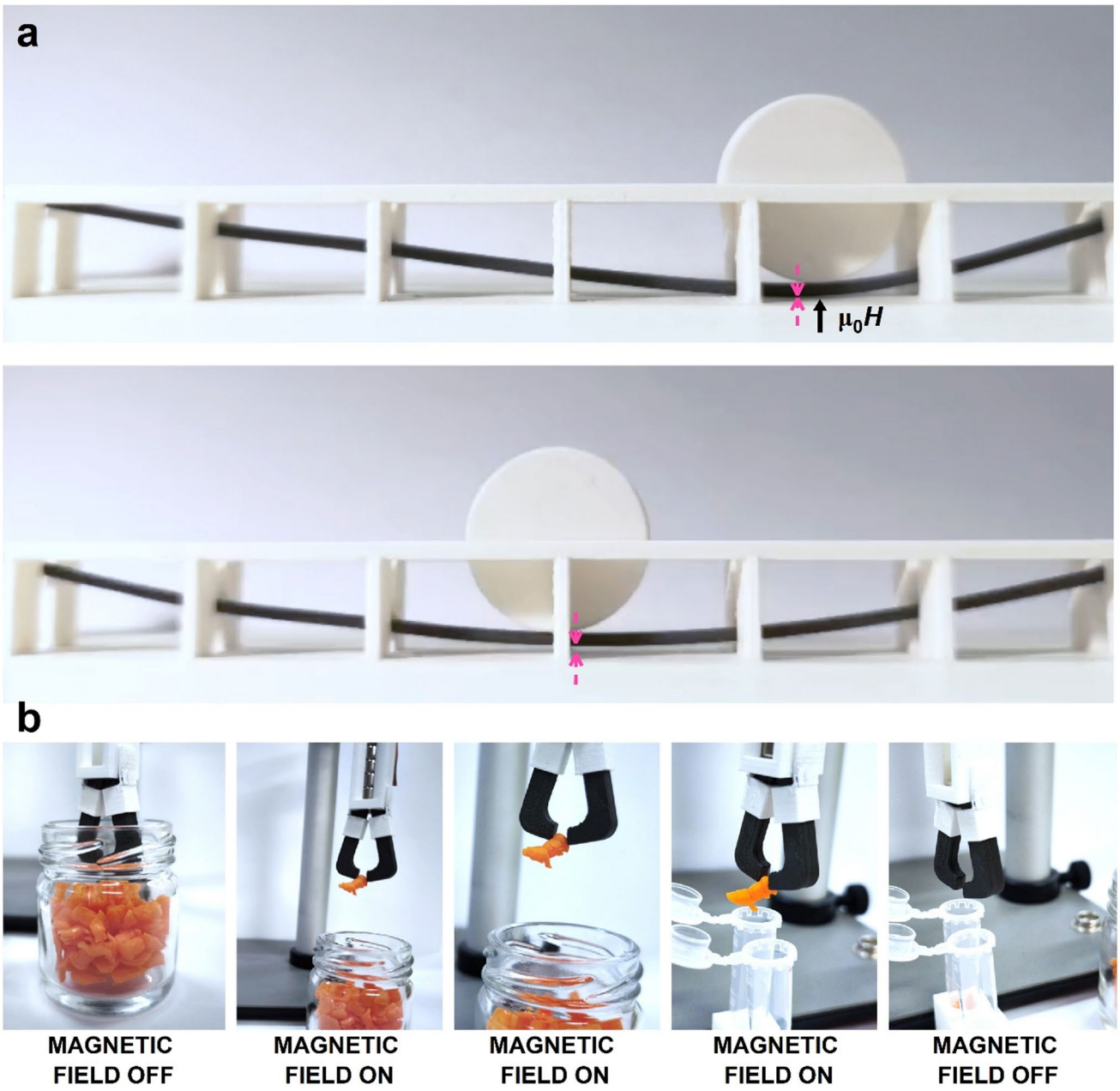


Fig. 5 Conceptual schematics illustrate potential applications enabled by soft magnetic composite filaments: (**a**) Photographs of the reversible magnetic actuation of the flexible magnetic composite under an external magnetic field, demonstrating field-induced deflection and full shape recovery upon field removal. See Supplementary Video 1. (**b**) Sequential photographs of magnetic manipulation using PETG+Fe-Nb-Si-B-Cu printed grippers: the gripper arms close around the object of interest (orange) and transfer it between containers under a magnetic field. It then releases the object as the magnetic field is removed. See Supplementary Video 2

under an external magnetic field, confirming that the high filler content (>50 wt%) preserves sufficient polymer compliance to enable field-responsive deformation. Upon removal of the magnetic field, the filament rapidly recovers its original configuration without visible fatigue (Fig. 5a, bottom panel), underscoring the mechanical resilience and reversible magneto-mechanical coupling of the composite network (see Supplementary Video 1 for further details). To further illustrate the versatility of the feedstock-engineered filaments, PETG+Fe-Nb-Si-B-Cu structures were printed into a magnetic gripper configuration (Fig. 5b). When subjected to an external magnetic field, the gripper arms close around the object of interest (shown in orange) and transfer it from a bottle into a vial, demonstrating manipulation driven solely by magnetic flux guidance. This behavior highlights that the composites are capable not only of responding to, but also of channeling, external fields for controlled actuation (see Supplementary Video 2 for further details). These demonstrations validate that the developed soft magnetic composites, combining high filler loadings, mechanical flexibility, and processing-induced magnetic softening, can be directly printed into 3D architectures that are simultaneously deformable and functional. Beyond these proof-of-concept systems, the materials establish a broader platform for magnetically reconfigurable devices, adaptive soft robotics, and field-responsive actuators, where motion and functionality are governed entirely by external magnetic stimuli rather than embedded wiring or heating elements.

4 Conclusions

The development of high-loading, soft magnetic composite filaments presented here demonstrates a substantial advance in the design and processing of functional feedstocks for additive manufacturing. This progress was made possible through a feedstock-engineering (encapsulation-extrusion) strategy, which enables the controlled integration of magnetic powders into diverse polymer matrices while preventing powder loss and ensuring homogeneous dispersion. By enclosing the powders within polymer capsules prior to extrusion, the approach eliminates air gaps, preserves the polymer–filler ratio, and allows extrusion of highly filled composites that remain continuous and printable. Using this methodology, we systematically compared three distinct polymer matrices with two classes of magnetic fillers, and identified the interplay between matrix chemistry, filler morphology, and processing conditions that achieves continuous, homogeneous filaments at loadings well above those of commercial analogs. The optimized fabrication route not only ensures uniform filler dispersion, as confirmed by electron microscopy, but also preserves the intrinsic amorphous or crystalline character of the magnetic powders, as evidenced by XRD, thus ensuring that desirable soft magnetic properties are retained during processing.

The contrasting performance of the polymer matrices provides mechanistic insight into the governing parameters of printability and flexibility. Among the matrices, PETG proved to be the most reliable for fabricating rigid, defect-free filaments at high loadings, while TPU uniquely enabled mechanically flexible, magnetically functional filaments. Such compliant composites combine magnetic responsiveness with resilience against fracture, opening opportunities in fields such as soft robotics, actuators, and flexible electronics, where mechanical compliance and functional integration are required simultaneously.

Beyond structural considerations, extrusion unexpectedly introduced a remarkable and reproducible effect: a decrease in coercivity relative to the raw powders, signifying a processing-induced magnetic softening. Rather than representing a deviation from expected behavior, this phenomenon reflects a functional enhancement arising from the extrusion process itself. The controlled thermal-mechanical environment during extrusion likely promotes stress relief. This observation transforms extrusion from a passive fabrication step into an active design parameter capable of tuning magnetic performance during filament fabrication.

The demonstrated feedstock-engineering methodology for good soft magnetic composite filament enables printed magnetic architectures that are flexible and also field-responsive. Demonstrations of magnetic actuation and flux-guided manipulation validate their potential in soft robotics, adaptive grippers, and electromagnetic control.

Taken together, these findings establish a versatile framework for engineering high-loading magnetic composites with structural integrity, mechanical compliance, and enhanced magnetic performance. By uniting high filler loading, filament continuity, and processing-induced functional gains, this work redefines the design space for magnetic composites in emerging applications ranging from soft robotics to field-responsive devices. While demonstrated for soft magnetic powders, the encapsulation-extrusion strategy could be readily extended to other functional particulates, such as dielectric, thermally conductive, or catalytic fillers, offering a generalizable blueprint for multifunctional composites that integrate structural properties with tailored performance within a single printed architecture.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s42114-026-01715-9.

Author contributions **G. Guerrero-Muñoz: ** Investigation, Data curing, Formal Analysis, Writing – reviewing & editing. **J Revuelta-Losada: ** Conceptualization, Investigation, Visualization, Formal Analysis, Writing – reviewing & editing. **J.Y. Law: ** Conceptualization, Investigation, Visualization, Writing – original draft, Supervi-

sion, Fund acquisition, Writing – reviewing & editing. **E. Guisado-Arenas: ** Investigation, Writing – reviewing & editing. **L.M. Moreno-Ramírez: ** Investigation, Formal Analysis, Writing – reviewing & editing. **M. Sanchez-Poncela: ** Investigation, Formal Analysis, Writing – reviewing & editing. **V. Franco: ** Conceptualization, Investigation, Visualization, Supervision, Fund acquisition, Writing – reviewing & editing.

Funding Funding for open access publishing: Universidad de Sevilla/CBUA. Work supported by grant PID2023-146047OB-I00 from AEI/10.13039/501100011033, US Air Force Office of Scientific Research (project FA8655-21-1-7044. JYL acknowledges EMERGIA fellowship (Ref. EMC21_00418) from Junta de Andalucía. EGA acknowledges FPU scholarship (Ref. FPU23/03426) from the Spanish MICIU.

Data availability Data available upon request from authors.

Declarations

Competing interests The authors declare no competing interests.

Conflict of interest The authors declare that they have no competing interests.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

References

1. Lazarus N, Bedair SS (2016) Improved power transfer to wearable systems through stretchable magnetic composites. Appl Phys A-Mater Sci Process 122:543. https://doi.org/10.1007/s00339-016-0067-y
2. Wu S, Hu W, Ze Q, Sitti M, Zhao R (2020) Multifunctional magnetic soft composites: a review. Multifunctional Mater 3:042003. https://doi.org/10.1088/2399-7532/abcb0c
3. Díaz-García Á, Law JY, Cota A, Bellido-Correa A, Ramírez-Rico J, Schäfer R, Franco V (2020) Novel procedure for laboratory scale production of composite functional filaments for additive manufacturing. Mater Today Commun 24:101049. https://doi.org/10.1016/j.mtcomm.2020.101049
4. Díaz-García Á, Law JY, Felix M, Guerrero A, Franco V (2022) Functional, thermal and rheological properties of polymer-based magnetic composite filaments for additive manufacturing. Mater Design 219:110806. https://doi.org/10.1016/j.matdes.2022.110806
5. Díaz-García Á, Revuelta J, Moreno-Ramírez LM, Law JY, Mayer C, Franco V (2022) Additive manufacturing of magnetocaloric $(La,Ce)(Fe,Mn,Si)_{13}$–H particles via polymer-based composite filaments. Compos Commun 35:101352. https://doi.org/10.1016/j.coco.2022.101352
6. Díaz-García Á, Law JY, Żrodowski Ł, Morończyk B, Wróblewski R, Franco V (2024) Polymer-based filaments with embedded magnetocaloric Ni-Mn-Ga Heusler alloy particles for additive manufacturing. Polym Compos 45:5333. https://doi.org/10.1002/pc.28130
7. Zhang Z, Heron JT, Pena-Francesch A (2023) Adaptive Magnetoactive Soft Composites for Modular and Reconfigurable Actuators. Adv Funct Mater 33:2215248. https://doi.org/10.1002/adfm.202215248
8. Bhatta A, McKenzie TF, DeFeo JR, Casey J, Yenugonda V, Mukaddaskhonov S, Toromani G, Pathak AK, Khan M (2025) Exploring the magnetic and magnetocaloric properties of PLA composite filaments of Mn0.5Fe0.5Ni0.95Cr0.05Si0.95Al0.05 for additive manufacturing. AIP Adv 15. https://doi.org/10.1063/9.0000942
9. Abbasi M, Váz P, Silva J, Martins P (2025) Head-to-Head Evaluation of FDM and SLA in Additive Manufacturing: Performance, Cost, and Environmental Perspectives. Appl Sci 15:2245. https://doi.org/10.3390/app15042245
10. Kafle A, Luis E, Silwal R, Pan HM, Shrestha PL, Bastola AK (2021) 3D/4D Printing of Polymers: Fused Deposition Modelling (FDM), Selective Laser Sintering (SLS), and Stereolithography (SLA). Polymers 13:3101. https://doi.org/10.3390/polym13183101
11. Monfared V, Bakhsheshi-Rad HR, Ramakrishna S, Razzaghi M, Berto F (2021) A Brief Review on Additive Manufacturing of Polymeric Composites and Nanocomposites. Micromachines 12:704. https://doi.org/10.3390/mi12060704
12. Giannotta N, Sala G, Bianchini C, Torreggiani A (2023) A Review of Additive Manufacturing of Soft Magnetic Materials in Electrical Machines. Machines 11:702. https://doi.org/10.3390/machines11070702
13. Zhou L, Miller J, Vezza J, Mayster M, Raffay M, Justice Q, Al Tamimi Z, Hansotte G, Sunkara LD, Bernat J (2024) Additive Manufacturing: A Comprehensive Review. Sensors-Basel 24:2668. https://doi.org/10.3390/s24092668
14. Huber C, Mitteramskogler G, Goertler M, Teliban I, Groenefeld M, Suess D (2020) Additive Manufactured Polymer-Bonded Isotropic NdFeB Magnets by Stereolithography and Their Comparison to Fused Filament Fabricated and Selective Laser Sintered Magnets. Materials 13:1916. https://doi.org/10.3390/ma13081916
15. Strong D, Kay M, Conner B, Wakefield T, Manogharan G (2018) Hybrid manufacturing – integrating traditional manufacturers with additive manufacturing (AM) supply chain. Additive Manuf 21:159. https://doi.org/10.1016/j.addma.2018.03.010
16. Jacob J, Pejak Simunec D, Kandjani AEZ, Trinchi A, Sola A (2024) A Review of Fused Filament Fabrication of Metal Parts (Metal FFF): Current Developments and Future Challenges. Technologies 12:267. https://doi.org/10.3390/technologies12120267
17. Bollig LM, Patton MV, Mowry GS, Nelson-Cheeseman BB (2017) Effects of 3-D Printed Structural Characteristics on Magnetic Properties. Ieee T Magn 53:1. https://doi.org/10.1109/TMAG.2017.2698034
18. Bollig LM, Hilpisch PJ, Mowry GS, Nelson-Cheeseman BB (2017) 3D printed magnetic polymer composite transformers. J Magn Magn Mater 442:97. https://doi.org/10.1016/j.jmmm.2017.06.070
19. Liu H, Wang F, Wu W, Dong X, Sang L (2023) 4D printing of mechanically robust PLA/TPU/Fe3O4 magneto-responsive shape memory polymers for smart structures. Compos Part B: Eng 248:110382. https://doi.org/10.1016/j.compositesb.2022.110382
20. Thomas JL, Andrew HM, Thomas AH, Brittany BN-C (2019) Dispersion of particulate in solvent cast magnetic thermoplastic polyurethane elastomer composites. AIMS Mater Sci 6:354. https://doi.org/10.3934/matersci.2019.3.354

21. Andrew HM, Thomas MC, Nathan AF, Thomas JL, John EW, Brittany BN-C (2019) Thermoplastic magnetic elastomer for fused filament fabrication. AIMS Mater Sci 6:363. https://doi.org/10.3934/matersci.2019.3.363
22. Rochman A, Delia S (2024) Additive Manufacturing of Thermoplastic Polyurethane Elastomers: A Review. Reference Module in Materials Science and Materials Engineering: Elsevier; 2024. https://doi.org/10.1016/B978-0-323-95486-0.00100-9
23. Su K-H, Su C-Y, Shih W-L, Lee F-T (2023) Improvement of the Thermal Conductivity and Mechanical Properties of 3D-Printed Polyurethane Composites by Incorporating Hydroxylated Boron Nitride Functional Fillers. Materials 16:356. https://doi.org/10.3390/ma16010356
24. Huber C, Cano S, Teliban I, Schuschnigg S, Groenefeld M, Suess D (2020) Polymer-bonded anisotropic SrFe12O19 filaments for fused filament fabrication. J Appl Phys 127. https://doi.org/10.1063/1.5139493
25. Slapnik J, Pulko I, Rudolf R, Anžel I, Brunčko M (2021) Fused filament fabrication of Nd-Fe-B bonded magnets: Comparison of PA12 and TPU matrices. Additive Manuf 38:101745. https://doi.org/10.1016/j.addma.2020.101745
26. Gonzalez-Gutierrez J, Cano S, Schuschnigg S, Kukla C, Sapkota J, Holzer C (2018) Additive Manufacturing of Metallic and Ceramic Components by the Material Extrusion of Highly-Filled Polymers: A Review and Future Perspectives. Materials 11:840. https://doi.org/10.3390/ma11050840
27. Waheed S, Cabot JM, Smejkal P, Farajikhah S, Sayyar S, Innis PC, Beirne S, Barnsley G, Lewis TW, Breadmore MC, Paull B (2019) Three-Dimensional Printing of Abrasive, Hard, and Thermally Conductive Synthetic Microdiamond–Polymer Composite Using Low-Cost Fused Deposition Modeling Printer. Acs Appl Mater Inter 11:4353. https://doi.org/10.1021/acsami.8b18232
28. Wang F-H, You C-Y, Tian N, Liu H-G, Zhang J, Zhu X-P (2024) 3D printing of soft magnetic materials: From printing to applications. J Alloy Compd 990:174486. https://doi.org/10.1016/j.jallcom.2024.174486
29. Khatri B, Lappe K, Noetzel D, Pursche K, Hanemann T (2018) A 3D-Printable Polymer-Metal Soft-Magnetic Functional Composite—Development and Characterization. Materials 11:189. https://doi.org/10.3390/ma11020189
30. Mazeeva A, Masaylo D, Razumov N, Konov G, Popovich A (2023) 3D Printing Technologies for Fabrication of Magnetic Materials Based on Metal–Polymer Composites: A Review. Materials 16:6928. https://doi.org/10.3390/ma16216928
31. ArcelorMittalPowders AdamIQ M300 - AISI 18Ni 300 | W. Nr 1.2709 | DIN X3NiCoMoTi 18-9-5 | UNS K93120 [cited 2025]. Available from: https://powders.arcelormittal.com/adamiq-m300-steel-powder
32. Sánchez-Poncela M, Van-Steenberge N, Gatti F, Rodríguez S (2021) inventorsMetal powder for Additive Manufacturing
33. ASTM_International., ASTM D790-17 (2017) Standard Test Methods for Flexural Properties of Unreinforced and Reinforced Plastics and Electrical Insulating Materials. ASTM International, West Conshohocken, PA
34. ASTM_International., ASTM D695-15 (2015) Standard Test Method for Compressive Properties of Rigid Plastics. ASTM International, West Conshohocken, PA
35. Omeire GC, Iwe MO, Nwosu JN (2013) Influence of Extrusion Variables on the Residence Time and Throughput of a Single Screw Extruder. Curr J Appl Sci Technol 3:277. https://doi.org/10.9734/BJAST/2014/2252
36. Darnal A, Shahid Z, Deshpande H, Kim J, Muliana A (2023) Tuning mechanical properties of 3D printed composites with PLA:TPU programmable filaments. Compos Struct 318:117075. https://doi.org/10.1016/j.compstruct.2023.117075
37. Krapež D, Jusufagić M, Obućina M, Kitek Kuzman M, Kariž M (2025) Effect of Filament Material and Printing Temperature on 3D Printing Extrusion Force. Appl Sci 15:2046. https://doi.org/10.3390/app15042046
38. Shanmugam V, Babu K, Kannan G, Mensah RA, Samantaray SK, Das O (2024) The thermal properties of FDM printed polymeric materials: A review. Polym Degrad Stab 228:110902. https://doi.org/10.1016/j.polymdegradstab.2024.110902
39. García Díez A, Pereira N, Tubio CR, Vilas-Vilela JL, Costa CM, Lanceros-Mendez S (2023) Magnetic Polymer Actuators with Self-Sensing Resistive Bending Response Based on Ternary Polymer Composites. ACS Appl Electron Mater 5:3426. https://doi.org/10.1021/acsaelm.3c00432
40. Pigliaru L, Rinaldi M, Ciccacci L, Norman A, Rohr T, Ghidini T, Nanni F (2020) 3D printing of high performance polymer-bonded PEEK-NdFeB magnetic composite materials. Funct Compos Mater 1:4. https://doi.org/10.1186/s42252-020-00006-w
41. Wu Z, Wang Q, Huang J, Yue Y, Chen D, Shi Y, Su B (2022) The soft NdFeB/Ecoflex composites for soft robot with a considerable magnetostimulated shrinkability. Compos Sci Technol 217:109129. https://doi.org/10.1016/j.compscitech.2021.109129
42. Arora N, Dua S, Singh VK, Singh SK, Senthilkumar T (2024) A comprehensive review on fillers and mechanical properties of 3D printed polymer composites. Mater Today Commun 40:109617. https://doi.org/10.1016/j.mtcomm.2024.109617
43. Franco V, Conde CF, Conde A (1998) Effect of the Si/B ratio on the magnetic anisotropy distribution of Fe73.5Si22.5-xBxCu1Nb3 (x = 7, 9, 16) alloys along nanocrystallization. J Appl Phys 84:5108. https://doi.org/10.1063/1.368807
44. Amirov A, Omelyanchik A, Murzin D, Kolesnikova V, Vorontsov S, Musov I, Musov K, Khashirova S, Rodionova V (2022) 3D Printing of PLA/Magnetic Ferrite Composites: Effect of Filler Particles on Magnetic Properties of Filament. Processes 10:2412. https://doi.org/10.3390/pr10112412
45. Gloeckle C, Konkol T, Jacobs O, Limberg W, Ebel T, Handge UA (2020) Processing of Highly Filled Polymer–Metal Feedstocks for Fused Filament Fabrication and the Production of Metallic Implants. Materials 13:4413. https://doi.org/10.3390/ma13194413

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.